

v44 model — experimental conditions: input parameters & where they sit in the proliferation parameter space

A Input parameters per condition

	SHH	Ptch1	fMYCN	p16	p18	p21	HHi	EZH2i	HU	CDK4/6i	onc?
GNP +SHH	0.5	1	1	0	0.4	1×	.	.	.	.	●
GNP +HHi	0.5	1	1	0	0.4	1×	✓	.	.	.	○
GNP +EZH2i	0.5	1	1	0	0.4	1×	.	✓	.	.	●
GNP serum-starve	0.5	1	1	0	0.4	1×	.	.	.	.	○
Ptch+/- (P7)	0.5	0.5	1.38	0.352	0.72	1.4×	.	.	.	.	●
MB	0.5	0.1	2.86	0.88	1.2	2×	.	.	.	.	●
MB +HHi	0.5	0.1	2.86	0.88	1.2	2×	✓	.	.	.	○
MB +HHi +EZH2i	0.5	0.1	2.86	0.88	1.2	2×	✓	✓	.	.	●
MB +CDK4/6i	0.5	0.1	2.86	0.88	1.2	2×	.	.	.	✓	○
MB +HU	0.5	0.1	2.86	0.88	1.2	2×	.	.	✓	.	●

context inputs (p16, p18, p21 = CDK-inhibitor tone)

drug switches

p16: competitive CDK4/6 brake (raises K_CdRb) · p21: CDK2 inhibitor (kSyP21, xbaseline) · serum-starve k_Cd_trans→0 · CDK4/6i kPhRbCd→0

B Parameter space: CyclinD1 vs the CDK-inhibitor brake

